

Diagrams that earn marks

Content

1. Labelling rules

A correct diagram is a set of marks in its own right, and a wrongly labelled one can lose the analysis marks that depend on it. Run through this list every time.

Axes. Label both. Micro: price (or costs / revenue) up the side, quantity along the bottom. Macro: price level and real GDP. Include units where you have them (£, %).

Every curve. Label each one (D, S, MC, AC, MR, AR) at its end, in the right place.

Key points. Mark the equilibrium or the profit-maximising point with a dot, and drop dashed lines to both axes.

Values on the axes. Write Q1 and P1, or the numbers from the extract, where the dashed lines meet the axes.

Shifts. Draw the new curve, label it D2 (or S2, AD2), and show the direction with an arrow. Label the new equilibrium Q2 and P2.

Areas. Shade and label areas that matter: supernormal profit, loss, consumer surplus, welfare loss.

Accuracy. MC cuts AC and AVC at their lowest points. MR starts on the price axis and is twice as steep as a straight-line AR. Use a pencil and ruler.

Size and space. Big enough to read, about a third to half a page, drawn where the question asks or next to your analysis.

2. Which diagram for which question

The Theme 3 diagrams that can be asked are listed below (from the Edexcel charts list). Look for the trigger words in the question.

If the question mentions...	Draw
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Profit maximisation, revenue maximisation, sales maximisation, objectives	AR, MR, MC and AC together. Profit max at $MC = MR$, revenue max at $MR = 0$, sales max at $AC = AR$.
Costs of production, diminishing returns	Short-run MC, AVC, AC (and AFC). MC cuts AVC and AC at their minima.
Economies or diseconomies of scale, minimum efficient scale	SRAC curves with the LRAC envelope, and MES at the bottom of the LRAC.
Revenue and elasticity	AR (demand), MR and TR. $MR = 0$ at the top of TR (unit elastic).
Shut-down, exit, staying in the market	AR against AVC (short run) or AC (long run). Shut down when AR falls below it.
Perfect competition, monopolistic competition, monopoly, natural monopoly	The short-run and long-run equilibrium for that structure, with the profit or loss area shaded.
Price discrimination	Third-degree: two separate markets with different elasticities, one MC line, and the $MR = MC$ output in each.
Allocative, productive or X-inefficiency	Allocative at $P = MC$, productive at the lowest point of AC, X-inefficiency as costs above the efficient AC curve.
Wages, employment, minimum wage, monopsony, trade unions	Labour market: demand (MRP) and supply, and the monopsony version with MRC. Show wage and employment before and after.

The full list, including Themes 1, 2 and 4, is in the Edexcel charts required document.

3. Integrating a diagram into your argument

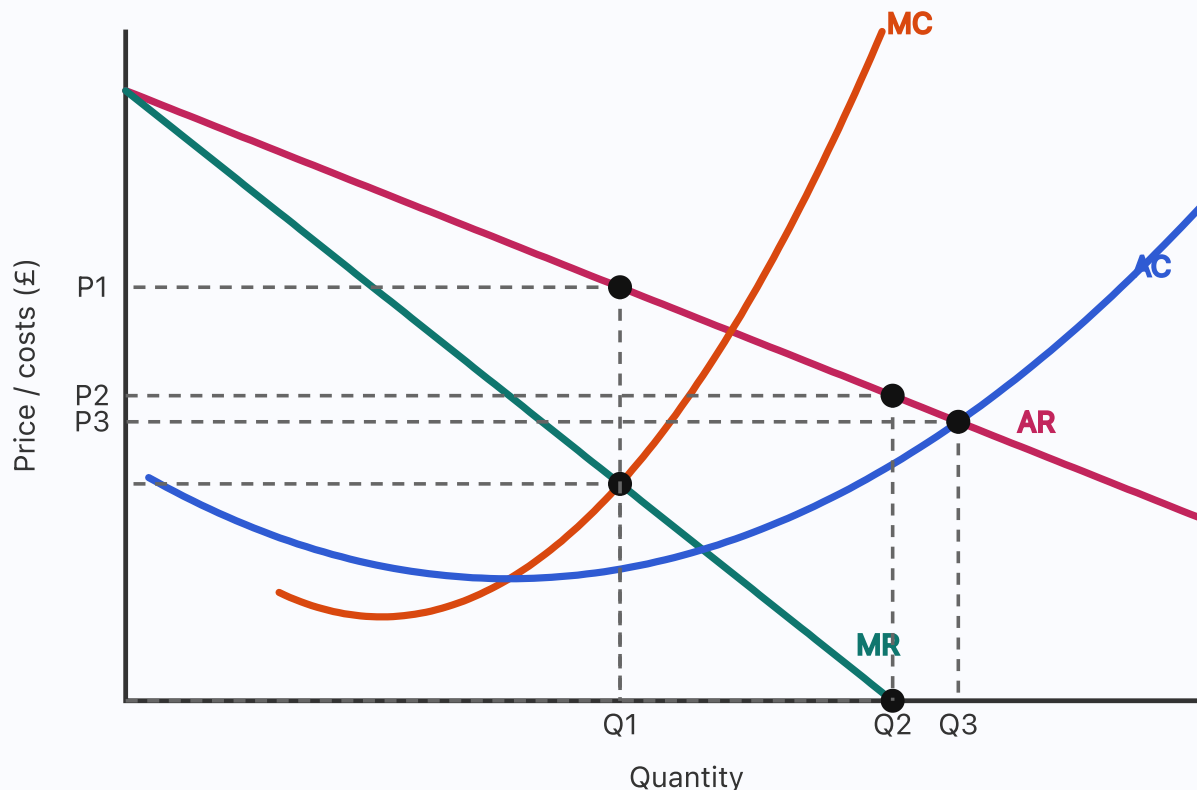
A diagram is evidence for a chain of reasoning, not a decoration. Examiners give credit when the diagram and the text tell the same story.

1. **Draw it at the point it is needed**, next to the paragraph it supports.
2. **Refer to it in the text.** "As shown in Figure 1, the profit-maximising output is Q1 where $MC = MR$."

3. **Use its labels in your chain.** Q1, P1 and C1 in the diagram should appear in the sentences.
4. **Explain what it shows** in words. Do not leave the examiner to work it out.
5. **Use it in evaluation.** Redraw or shift a curve to show the "maybe not" case.

Worked diagram: three objectives on one graph

Step through it. Notice how each objective picks a different output.



1. Demand (**AR**) slopes down. **MR** starts at the same point on the price axis and is twice as steep.
2. Add **MC** and **AC**. MC cuts AC at its lowest point.
3. **Profit maximisation:** $MC = MR$ at Q_1 . Read up to AR for the price, P_1 .
4. **Revenue maximisation:** $MR = 0$ at Q_2 . Total revenue is at its peak, output is higher and price lower than at Q_1 .
5. **Sales maximisation:** $AC = AR$ at Q_3 , the highest output that still earns normal profit.
6. Order of output: $Q_1 < Q_2 < Q_3$. Order of price: $P_1 > P_2 > P_3$. A firm moving from profit maximisation to sales maximisation trades profit for volume.

Common diagram mistakes

Unlabelled axes or curves

Without labels, the examiner cannot give credit for the diagram.

MC crossing AC at the wrong place

MC must cut AC at its lowest point.

MR not steeper than AR

For a downward-sloping demand curve, MR lies below and is twice as steep.

Diagram not used in the text

The diagram is drawn, but the paragraph never refers to it.

Too small or too messy

A cramped diagram hides mistakes and cannot be read quickly.